ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Integration

Integrate by reversing differentiation, then use constants or limits.

INTEGRATION · P1 · 13

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 POWER: Integrate Powers
TRIGGER *integrate polynomial, roots, fractions*

02 EXPAND: Expand Before Integrating
TRIGGER *brackets or fractions before integration*

03 CONST: Find The Constant
TRIGGER *curve passes through a point*

04 AREA: Area Under Curve
TRIGGER *area bounded by curve and axis*

BANK EVIDENCE Local pages: 13 problem, 26 answer, 7 notes. Website primary entries: 32.

01 POWER: Integrate Powers

TRIGGER *integrate polynomial, roots, fractions*

BECOMES Increase the power by 1 and divide by the new power.

FIRST LINE TO WRITE

$$\int ax^n\,dx = \frac{a}{n+1}x^{n+1} + c$$

SIMPLEST STRATEGY

1. **P**ut terms into power form.
2. **O**ne is added to each power.
3. **W**rite coefficient divided by new power.
4. **E**nd with $+c$.
5. **R**ewrite neatly.

WORKED MODEL

$$\int 6x^2\,dx = 2x^3 + c.$$

HARD VARIANTS

1. Rewrite roots and denominators as powers before integrating.
2. Do not use the power rule for x^{-1}; that special case is outside this Pure 1 move.
3. Include $+c$ unless limits are present.

BOTTOM LINE

The $+c$ belongs to indefinite integration.

Practice 01

SAME IDEA Use POWER: Integrate Powers.

QUESTION

Integrate $4x^3 - 3x^{-2}$.

YOUR SOLUTION

02 EXPAND: Expand Before Integrating

TRIGGER *brackets or fractions before integration*

BECOMES Make each term a simple power.

FIRST LINE TO WRITE

$$\frac{x^2+3x}{\sqrt{x}} = x^{3/2} + 3x^{1/2}$$

SIMPLEST STRATEGY

1 **E**xpand brackets.
2 **X**press roots/fractions as powers.
3 **P**ut each term separately.
4 **A**pply power rule.
5 **N**eaten coefficients.
6 **D**on't forget $+c$.

WORKED MODEL

$$\frac{(x+2)^2}{\sqrt{x}} = x^{3/2} + 4x^{1/2} + 4x^{-1/2}.$$

HARD VARIANTS

1 Expand brackets and split fractions before integrating term by term.
2 If the numerator is a product, multiply out first unless a substitution is clearly intended.
3 Keep fractional powers exact until the final simplification.

BOTTOM LINE

Integration starts after simplification.

Practice 02

SAME IDEA Use EXPAND: Expand Before Integrating.

QUESTION

> Rewrite and integrate $\frac{(x+1)^2}{\sqrt{x}}$.

YOUR SOLUTION

03 CONST: Find The Constant

TRIGGER *curve passes through a point*

BECOMES Integrate first, then substitute the point.

FIRST LINE TO WRITE

$$y = F(x) + c, \quad (x, y) \text{ known}$$

SIMPLEST STRATEGY

1 **C**alculate the integral.
2 **O**btain $+c$.
3 **N**ame the point.
4 **S**ubstitute x, y.
5 **T**ell final function.

WORKED MODEL

$$y = x^2 + c,\ (3, 10) \Rightarrow 10 = 9 + c,\ c = 1.$$

HARD VARIANTS

1 Integrate first, then use the point to find c.
2 If given gradient and point, do not substitute into the derivative.
3 For two conditions, use them after the general function is written.

BOTTOM LINE

The constant is found from a point on the curve.

Practice 03

SAME IDEA Use CONST: Find The Constant.

QUESTION

Given $f'(x) = 2x$ and $f(4) = 9$, find $f(x)$.

YOUR SOLUTION

04 AREA: Area Under Curve

TRIGGER *area bounded by curve and axis*

BECOMES Integrate between the limits.

FIRST LINE TO WRITE

$$\text{Area} = \int_a^b y\,dx$$

SIMPLEST STRATEGY

1. **A**ssign lower and upper limits.
2. **R**ead whether area needs splitting.
3. **E**valuate the integral.
4. **A**nswer positive area.

WORKED MODEL

$$\int_0^2 x^2 dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}.$$

HARD VARIANTS

1. If the curve crosses the axis, split the integral and take positive area.
2. For area between two curves, integrate top minus bottom.
3. Use exact limits and state square units if the context needs them.

BOTTOM LINE

Area is positive; split if curve crosses the axis.

Practice 04

SAME IDEA Use AREA: Area Under Curve.

QUESTION

Find the area under $y = x + 1$ from $x = 0$ to $x = 3$.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
integrate polynomial, roots, fractions	$\int ax^n\,dx = \frac{a}{n+1}x^{n+1} + c$
brackets or fractions before integration	$\frac{x^2+3x}{\sqrt{x}} = x^{3/2} + 3x^{1/2}$
curve passes through a point	$y = F(x) + c, \quad (x, y)$ known
area bounded by curve and axis	Area $= \int_a^b y\,dx$